

An Orientation on

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UG MINOR PROJECT



Integrated MCA + BCA + BCA Data Science

Main Objective

Make the Minor Project
SERIOUSLY FUN !!



Why Minor Project?

- The goal of this Minor Project course is to help the students apply the theories and important tools they studied in this program to practice Computer Science/Data Science to mobilize the students and improve their skills for the next semester's Major Project course.

CREDITS - 4

Total Marks - 100

Road Map of Project Work

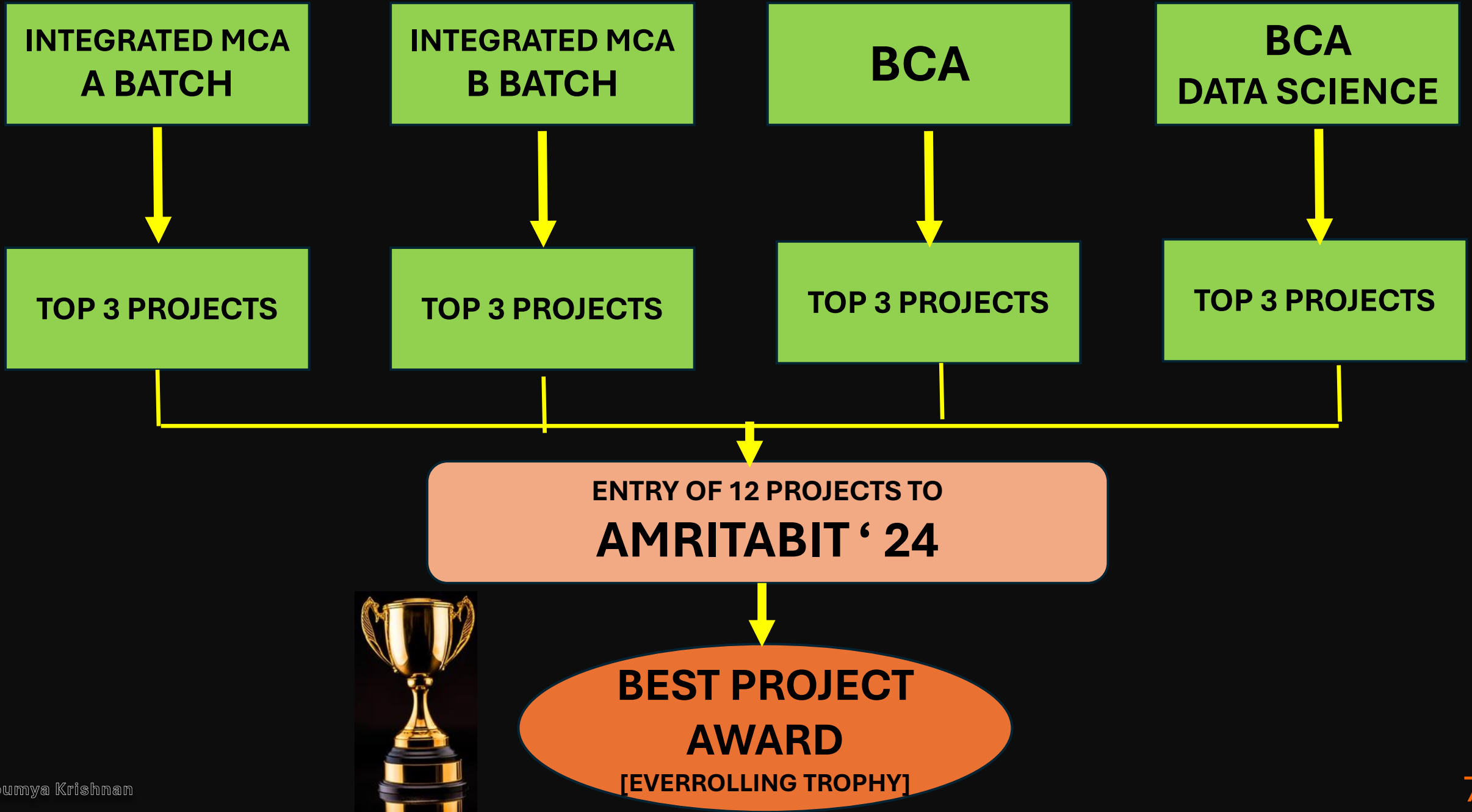


**Best
Project
Award**



Entry to
AMRITABIT
&

Attainment of Ever rolling Trophy for
BEST PROJECT



Steps in Detail

1. Form Groups

- Form a Group / Team containing FOUR members.
- Choose a Group Leader.

2. Choose a Relevant Topic

- Select 3 – 4 topics of your
 - ✓ Interest
 - ✓ Real & Relevant
 - ✓ Innovative
 - ✓ Useful for Society & Yourself

How to choose a Project Topic?

- Selecting the right topic, information source and proper guidance is most important for completing the project successfully.
- Choose a problem statement that needs a solution in real life. Create a software/ website/ app or any related project to establish the solution for the objective.
- Always choose a topic that interests you. Remember that this project will have an impact on your portfolio. So, choose a topic that will align with your career goals.

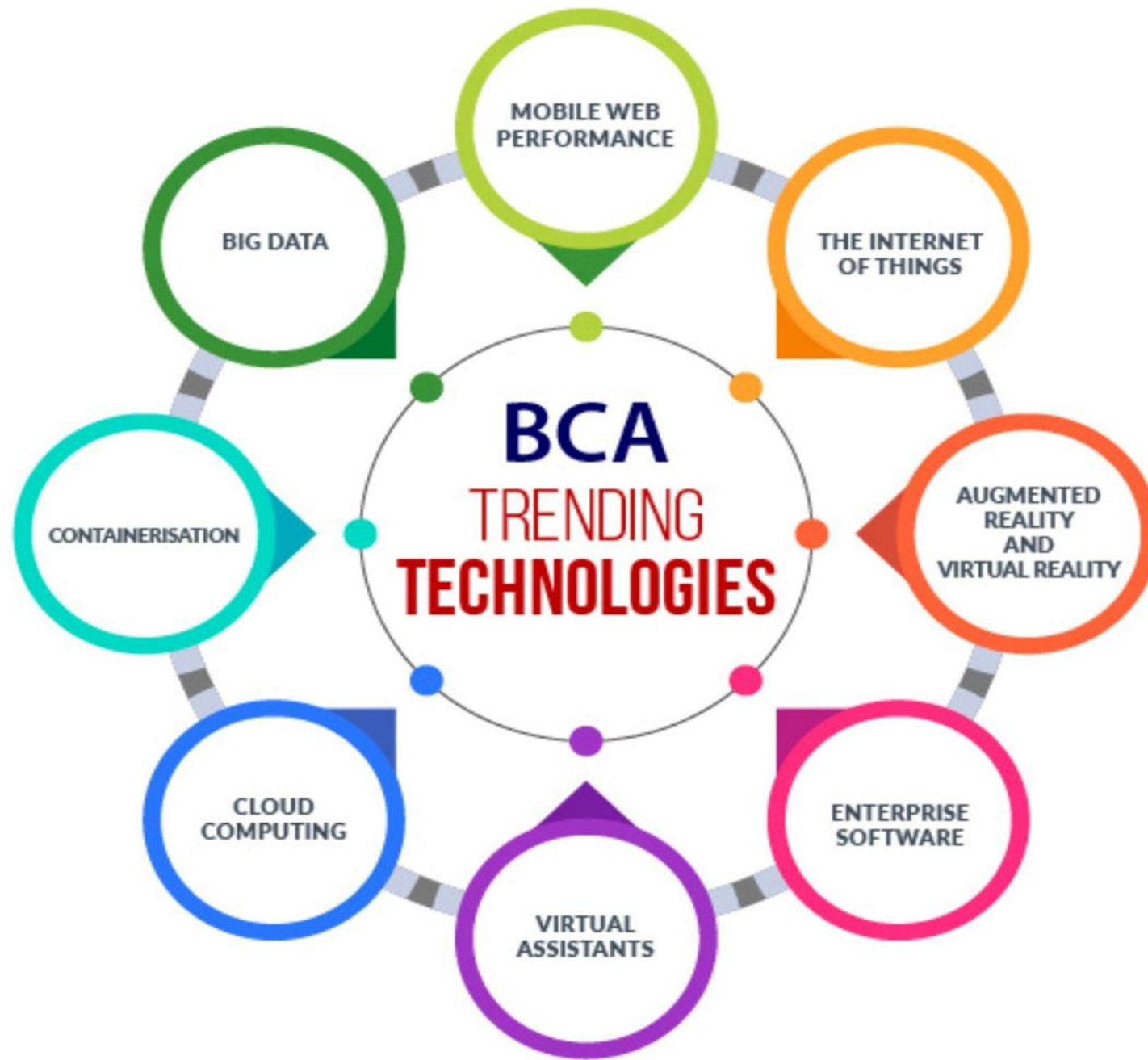
The important criteria for choosing a project are:-

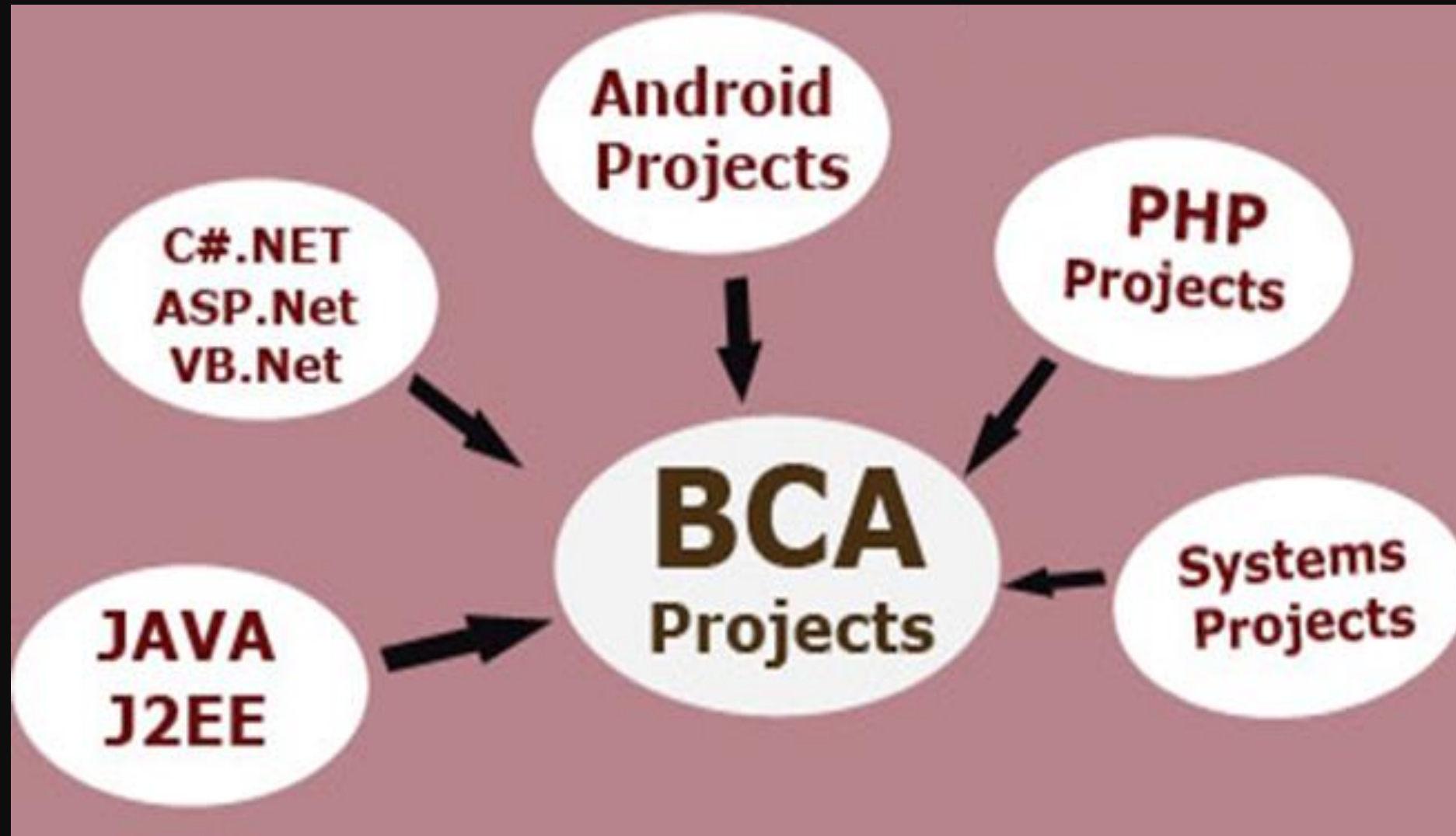
- Realistic
- Align with Future Plans & Goals.
- Identifying the constraints.
- Proper Planning

Knowing & Choosing the Types of Software Development Projects

Types of Software Developments

1. *Management Software Development*
2. Desktop Application Development
3. *Enterprise Software Development*
4. *Web Application Development*
5. *Mobile Application Development*
6. Game Development Software
7. Security Software System Development
8. Embedded Software Development
9. Software Tools Development
10. Data Science / ML Projects
11. Database Software Development





40 PROJECT IDEAS FOR WEB DEVELOPER

1. Calculator



2. Quiz App



3. Rock Paper Scissors



4. Note App



5. Stopwatch App



6. QR Code Reader



7. Weather App



8. Ecommerce Website

9. Landing Page

10. Password Generator

11. Tic Tac Toe Game

12. Link Shortener Website



13. Portfolio Website



14. Drawing App



15. Food Order Website

16. Meme Generator



17. Movie App

18. Chat App



19. Twitter Clone

20. Survey App

21. E-Book Site



22. Instagram Clone



23. Whatsapp Clone



24. Netflix Clone

25. File Sharing App



26. Parallax Website

27. Job Search App:

28. Pinterest Clone



29. Dating App

30. Social Media Dashboard

31. Tracker App



32. Memory App



33. Giphy Clone

34. User Activity Tracker



35. Stock-Trading App

36. Chess Game



37. Music Player



38. To-Do List App

39. Random User API



40. Typing Speed Test

Some Sample Project Topics

- ✓ Mini Search Engine
- ✓ Sentiment Analysis system
- ✓ Music organizer
- ✓ Chatbot
- ✓ Social media apps.
- ✓ Sales forecasting system for e-commerce food products
- ✓ System to detect fake logos online
- ✓ Wishlist products price comparison

- ✓ Project for advanced video surveillance
- ✓ Mobile quiz through Wi-Fi
- ✓ Secure ATM using card scanning plus OTP
- ✓ Stream analysis for career choice aptitude tests
- ✓ College bot project
- ✓ Online Quiz Application
- ✓ Task Manager
- ✓ Inventory Management System
- ✓ Recipe Finder
- ✓ Project on image mining

- ✓ Smart health advisory system
- ✓ Contact Management System
- ✓ Weather App
- ✓ E-commerce Website
- ✓ Resume Builder
- ✓ Chat Application
- ✓ Movie Recommendation System

Data Science / ML Project Samples

☐ Healthcare

- ✓ Disease Prediction Using Machine Learning
- ✓ Heart Disease Prediction Using Logistic Regression
- ✓ Prediction of Wine type using Deep Learning
- ✓ Parkinson's Disease Prediction using Machine Learning in Python
- ✓ Breast Cancer Wisconsin Diagnosis using Logistic Regression
- ✓ Cancer cell classification using Scikit-learn
- ✓ Autism Prediction using Machine Learning

❑ Finance and Economics

- ✓ Credit Card Fraud Detection
- ✓ Bitcoin Price Prediction using Machine Learning in Python
- ✓ Online Payment Fraud Detection using Machine Learning in Python
- ✓ Stock Price Prediction using Machine Learning in Python
- ✓ Stock Price Prediction Project using TensorFlow
- ✓ Predicting Stock Price Direction using Support Vector Machines
- ✓ Share Price Forecasting Using Facebook Prophet

❑ Food and Beverage

- ✓ Wine Quality Prediction

❑ Retail and Commerce

- ✓ Sales Forecast Prediction
- ✓ Customer Segmentation using Unsupervised Machine Learning
- ✓ Analyzing the selling price of used cars
- ✓ Box Office Revenue Prediction Using Linear Regression in ML
- ✓ Flipkart Reviews Sentiment Analysis
- ✓ Loan Approval Prediction using Machine Learning

❑ Sports

- ✓ IPL Score Prediction Using Deep Learning

❑ Health and Fitness

- ✓ Calories Burnt Prediction using Machine Learning
- ✓ Medical Insurance Price Prediction using Machine

❑ Transportation and Traffic

- ✓ Vehicle Count Prediction From Sensor Data
- ✓ Ola Bike Ride Request Forecast using ML

3 – Meet your Guide & Fix a Topic

- Select the most realistic, relevant, and innovative topic of your interest with the approval of your guide.

4. THOROUGH SYSTEM STUDY

- System study involves :

- ✓ Study of an Existing System,
- ✓ Documenting of existing system,
- ✓ Identifying current deficiencies, and
- ✓ Establishing new objectives and goals.

5. DECIDE ON PROJECT MODULES, FRONTEND & BACKEND TOOLS

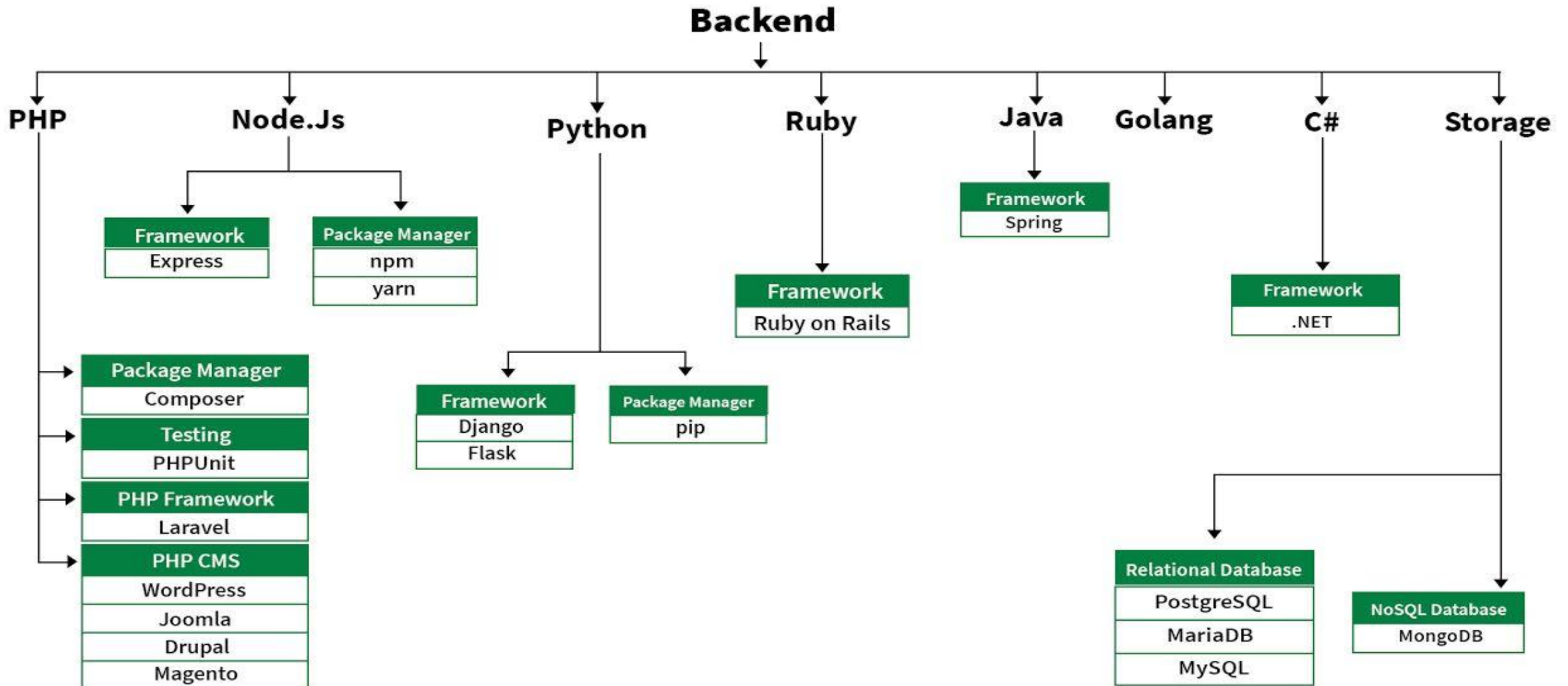
- Correctly identify the SCOPE of the project.
- Identify the different MODULES you plan to complete.
- Decide upon your frontend and backend tool.

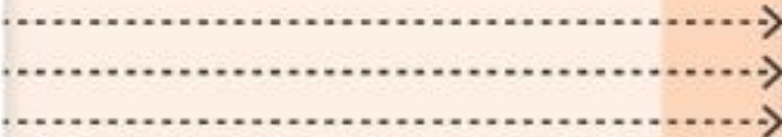
Top 20 Frontend Technologies

- ✓ HTML5 Boilerplate
- ✓ React JS
- ✓ Angular JS
- ✓ Vue JS
- ✓ Next JS
- ✓ Gatsby JS
- ✓ React Native
- ✓ Flutter
- ✓ Monorepo
- ✓ Micro frontends
- ✓ Three JS

- ✓ Remix
- ✓ Headless CMS
- ✓ GraphQL
- ✓ PWA
- ✓ Design System
- ✓ NPM
- ✓ Ionic
- ✓ Bootstrap
- ✓ Webpack

Backend Technologies





Frontend

The process of adding functionality to graphical user interface elements to create web experiences.

Focus

How websites and applications look

Languages

CSS

HTML

JavaScript

Technologies

Code editors (i.e., VS Code)

Version control software (i.e., Git)

Frontend frameworks (i.e., React)

Backend

The process of creating the logic, databases, and data structures that power web experiences.

Focus

How websites and applications work

Languages

Python

PHP

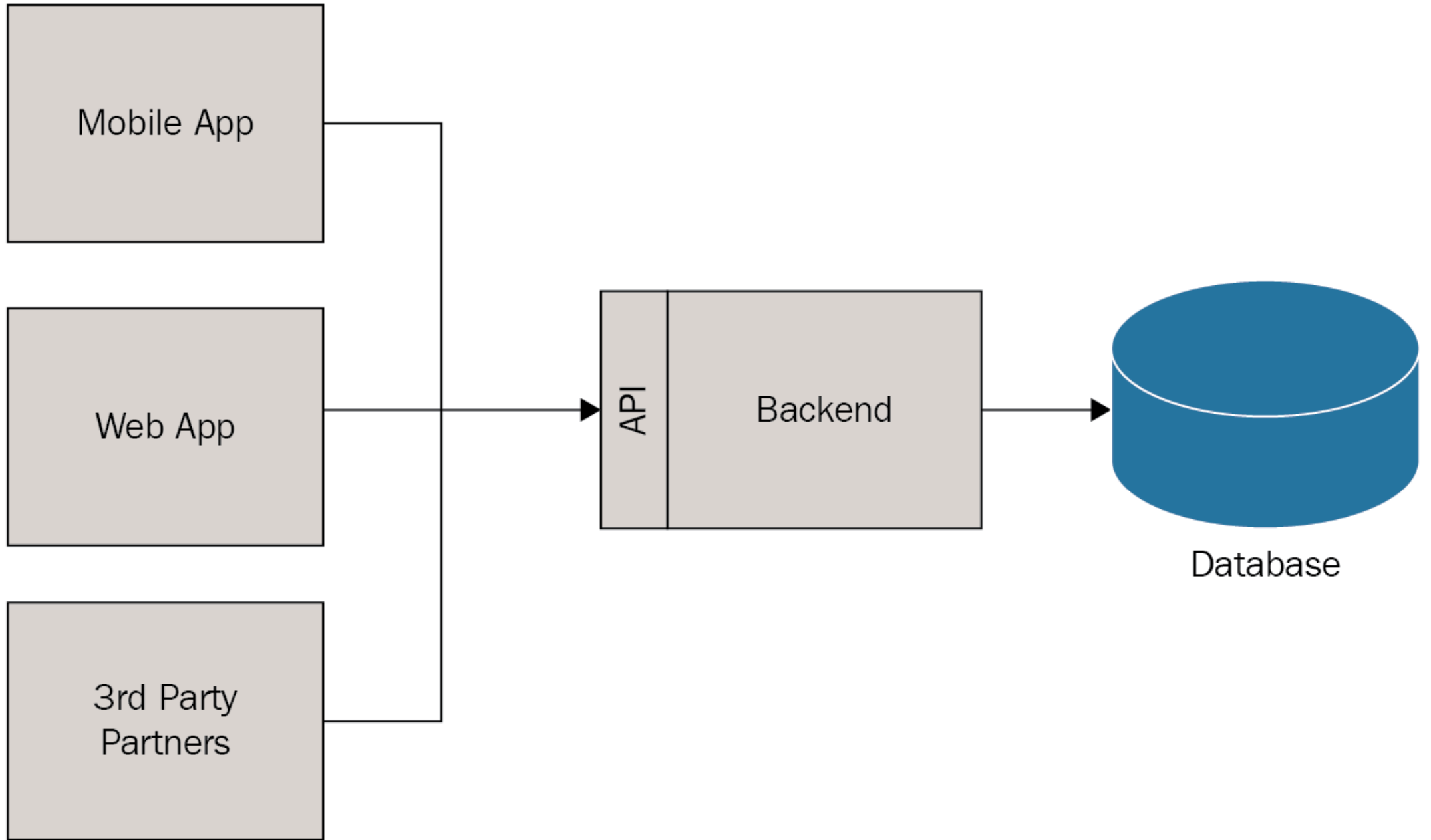
Ruby on Rails

Technologies

Servers (i.e., Apache HTTP Server)

Security protocols (i.e., SSL)

Backend frameworks (i.e., MySQL)



6. Prepare & Submit Synopsis

- A mini document containing a detailed description of your project must be prepared and submitted to your guide and get the approval for proceeding forward. The synopsis may have a format to follow.

Synopsis Format

Project Title

Name of the Students & Guide

1. INTRODUCTION
 - 1.1 About the System
 - 1.2 Need for the System
2. BACKGROUND STUDY
 - 2.1 Existing System & Demerits
 - 2.2 Proposed System & Merits
3. NEW SYSTEM DESCRIPTION
 - 3.1 Modules of the System
 - 3.2 Users of the System
 - 3.3 Future Scope
4. SOFTWARE & HARDWARE REQUIREMENTS
5. CONCLUSION

7. Prepare ERD, DFD/AD, Table Design

- If dealing with DB projects, prepare
 - ✓ Entity Relationship Diagram
 - ✓ Data Flow Diagram
 - ✓ Table Design with Normalization
- If dealing with non-DB Projects, prepare
 - ✓ Activity Diagram
- Data Science / ML Projects
 - ✓ Literature Review Reports
 - ✓ Activity Diagram
 - ✓ Detailing of Dataset

8. Appear for REVIEW 1

- PPT presentation describing all the work till date.
- Submit Synopsis + other documents (as per Step 7).
- **Review 1 evaluation – 25 Marks**
- Mandatory Consultation with your Guide (5 MARKS).
- Marks are transparent & shared with Students.
- Evaluation done by assigned Review Panels.
- Each student will be evaluated based on their contribution to the project & performance during the review.

9. PREPARE FRONTEND, BACKEND & ARCHITECTURAL DESIGN

- User Interface Design
- Database design with the business layer (if any)
- Architectural design of the system.

10. Start Coding

- You can start your coding by choosing relevant IDEs, and suitable programming languages along with identifying database (backend) software.
- Data Science Students can start implementing the system by choosing relevant ML methodologies with datasets.

11. Appear for REVIEW 2

- 40% - 50% completion of coding is mandatory.
- PPT presentation describing all the work till date.
- Submit document involving Steps 8 & 9.
- If possible, show implementation done till date.
- **Review 2 evaluation – 25 Marks**
- Mandatory Consultation with your Guide (5 MARKS).
- Marks are transparent & shared with Students.
- Evaluation done by assigned Review Panels.
- Each student will be evaluated based on their contribution to the project & performance during the review.

12. Complete Coding + Documentation

- Mandatory completion with Coding, manual Testing & Validation.
- Prepare the documentation using MS WORD / LATEX, covering the entire details of the project.
- Format of Documentation will be shared on time.
- Submission & getting approval from GUIDE.

13. Prepare your Record Books

- Updates and corrections to be done based on the Guide's review comments.
- Record Book completion including entire details of project work + sample code.
- Ready the hard copies based on Instructions from Project Coordinators.

14. Appear for Final REVIEW

- Execution of the Project
- Evaluation of the Record Books.
- Viva-voce based on Project.
- Evaluation done by assigned Review Panels.
- Each student will be evaluated based on their contribution to the project & performance during the review.
- **Final Review – 50 Marks**

PROJECT COMPLETION MARKED !!

15. Entry to AMRITABIT'24

- Best 12 Projects goes in for AmritaBit'24 (*a project exhibition conducted every semester*)
- One Best Project will be awarded with the ever-rolling Trophy.



IMPORTANT DATES

- Group / Team Formulation – 20/07/2024 (Saturday)
- Project Title Finalization – 25/07/2024 (Thursday)
- Synopsis Submission – 05/08/2024 (Monday)
- REVIEW 1 – 17/08/2024 (Saturday)
- REVIEW 2 – 05/10/2024 (Saturday)
- FINAL REVIEW – 02/11/2024 (Saturday)
- AMRITABIT'24 – 2nd / 3rd week of November 2024

Doubts ???

Thank You !!!